# Phase 1 Analysis: Impact of Hardware Constraints on IRS-Assisted Secure Sensing in Industry 5.0

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***Abstract*—This report establishes the theoretical and practical foundation for deploying Intelligent Reflecting Surfaces (IRS) in Industry 5.0 smart factories. We contrast the standard "ideal" IRS assumptions found in literature with the realistic hardware constraints of low-cost, 1-bit programmable metasurfaces. Specifically, we analyze the impact of phase quantization and amplitude-phase coupling (resonance loss) on signal integrity. This analysis defines the core engineering challenge for secure spectrum sensing in dense industrial environments.**

***Index Terms*—Industry 5.0, Intelligent Reflecting Surface, 1-Bit Quantization, Smart Manufacturing, Physical Layer Security.**

## I. Introduction: The Industry 5.0 Context

Industry 5.0 represents a paradigm shift from purely automated mass production to a human-centric, resilient, and sustainable model [1]. In this environment, humans work alongside collaborative robots ("cobots"), requiring wireless networks that are exceptionally reliable.

### *A. The Problem: Signal Blockage*

Smart factories are dense with metallic structures, moving robots, and heavy machinery. These obstacles create severe Non-Line-of-Sight (NLoS) conditions and electromagnetic interference [1]. This results in "dead zones" where Ultra-Reliable Low Latency Communication (URLLC) fails, endangering human-robot collaboration.

### *B. The Solution: IRS*

The Intelligent Reflecting Surface (IRS) acts as a "programmable mirror." By installing these surfaces on walls, factory operators can create "virtual" line-of-sight paths to bypass obstacles, ensuring continuous connectivity (targeting $10^{-9}$ error rates) even when direct paths are blocked [1].

## II. The "Ideal" Assumption vs. Reality

### *A. Theoretical View (The Ideal Model)*

Most theoretical research assumes an ideal IRS architecture to simplify optimization problems. In standard simulations, the following simplifications are often made [2]:

- **Continuous Phase Shifting:** It is assumed that each unit cell can shift the phase of an incoming signal by any arbitrary angle $\theta \in [0, 2\pi]$.
- **Perfect Reflection:** It is assumed that the reflection amplitude is always 1 (100% energy reflected), regardless of the phase shift applied.
- **Decoupled Control:** Phase and amplitude are treated as independent variables; changing the phase does not affect signal strength.

### *B. Hardware Reality (Practical View)*

Real hardware is far more constrained than theory suggests. Based on the 32.8 GHz mmWave prototype detailed in [2], we must model two critical imperfections for this project:

*1) 1-Bit Quantization:* Low-cost industrial IRS units typically do not support continuous phase shifting. They utilize PIN Diodes (e.g., AlGaAs) that function as simple binary switches.

- **Constraint:** The elements have only two discrete states: State 0 (OFF, $\approx 0°$) and State 1 (ON, $\approx 180°$) [2].
- **Impact:** This prevents perfect beam alignment. We are limited to a coarse alignment with significant quantization error, resulting in a wider, weaker reflected beam compared to theoretical predictions.

*2) Amplitude-Phase Coupling (The "Dip"):* In realistic implementations, shifting the phase causes a loss in signal power, known as Amplitude-Phase Coupling.

- **Mechanism:** To achieve the required $180°$ phase shift, the unit cell (dumbbell shape) must resonate. At the operating frequency of 32.8 GHz, this resonance causes a sharp drop in reflection amplitude [2].
- **Observed Behavior:** In the OFF state, reflection amplitude is high ($> 0.9$) but the phase is uncontrolled. In the ON state, the phase is correct ($180°$), but the reflection amplitude drops to $\approx 0.2-0.4$ due to resonance absorption [2].

## III. Implication for Project Simulation

This contrast defines the core engineering challenge of this project. Standard security algorithms assume we can fine-tune phases to maximize the user's signal while nulling the eavesdropper. In our simulation, we must determine if secure sensing is still possible when constrained to just two phase states ($0°/180°$) and suffering signal loss (amplitude dip) every time we switch a bit.

The subsequent phases of this project will implement a channel model that explicitly accounts for these binary constraints and power losses to evaluate the robustness of decentralized learning algorithms [3], [4].

## REFERENCES


[1] M. Noor-A-Rahim, et al., "Towards Industry 5.0: Intelligent Reflecting Surface (IRS) in Smart Manufacturing," *arXiv:2201.02214*, Jun. 2022.

[2] H. T. S. ALRikabi, et al., "Design and experimental evaluation of a reconfigurable intelligent surface for wireless applications," *Results in Engineering*, vol. 26, 104694, 2025.

[3] Z. Wang, et al., "IRS-Enhanced Spectrum Sensing and Secure Transmission in Cognitive Radio Networks," *IEEE Trans. Wireless Commun.*, vol. 23, no. 8, Aug. 2024.

[4] M. Rossanese, et al., "Open Experimental Measurements of Sub-6GHz Reconfigurable Intelligent Surfaces," *IEEE MultiMedia*, 2024.